ConceptFormer-2: Token-Efficient Grounding of Frozen Instruction-Tuned Language Models in Knowledge Graphs

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Abstract

Retrieval grounds language models in external knowledge, but it pays for this grounding on every query with hundreds of context tokens. ConceptFormer-2 instead compresses an entity's Wikidata neighborhood into k continuous concept tokens that a frozen, instruction-tuned language model consumes in place of retrieved text. A small resampler builds these tokens from the frozen model's own label embeddings, so an unseen entity requires one forward pass and no new parameters. Training needs no labels: the frozen model reading k concept tokens is distilled toward the same frozen model reading the facts as text. On PopQA, eight concept tokens lift a frozen Qwen3-0.6B from 0.10 to 0.48 exact match on unseen entities, an accuracy that text-based baselines reach only at about five to six times the token cost. Accuracy grows with token budget and with training data. Larger frozen backbones gain less from injection, a decline that replicates across the Qwen3 and Gemma-3 families; in Qwen the shortfall is non-monotone in model size, with a mid-size valley. In both families, run-to-run variance at the largest corpus overtakes the effects under study. Counterfactual probes confirm that answers follow the injected graph, and the encoder transfers zero-shot to MetaQA, raising accuracy from 0.06 to 0.34 on a knowledge graph it never saw in training, and reaching a similar ceiling on a second foreign graph. This transfer holds across graphs but stops at the label language: an English-trained encoder does not carry over to German-labeled graphs. All corpora, checkpoints, and per-item evaluation outputs are released.

1 Introduction

Instruction-tuned language models answer questions about popular entities from their weights, but they fail on the long tail, a deficit that persists even in frontier models [42, 43]. Retrieval-augmented generation [44] restores factuality at a recurring cost. A single entity's verbalized Wikidata neighborhood occupies a median of roughly 100 context tokens in our corpora, and on every query this text competes with the user's actual task for the context window.

ConceptFormer [1] proposed a third option: compress the knowledge-graph neighborhood itself into a few continuous tokens that a frozen language model consumes natively. The original system was validated on GPT-2 with rank-based metrics for next-token factual recall, and it left the central questions open. Rank-based scores cannot tell whether a model would actually generate a fact, because a gold answer among the top logits can still lose the greedy decode to a competing wrong one. Whether the model reads the injected structure or treats it as an opaque signal was never tested. The expectation that injection, like soft prompting [53], would benefit from larger frozen models also remained a conjecture. ConceptFormer-2 measures all three: whether latent injection holds under generative, exact-match question answering with a modern conversational model, whether the model reads the injected graph, and what governs the value of this channel among the token budget, the amount of knowledge the encoder is trained on, and the size of the frozen model.

ConceptFormer-2 addresses these questions with a new design. It makes three changes. The featurizer represents each graph edge with the frozen model's own label embeddings instead of v1's per-entity lookup table, which makes the encoder inductive: an unseen entity needs only its labels. The training objective is label-free. The frozen model reading k concept tokens is distilled toward the same frozen model reading the facts as text, with a full-vocabulary KL loss and no QA labels in the loss. The gold answer is used only to construct the teacher's context and to tier the corpus, never as a training target. Finally, the encoder's output projection is zero-initialized, so training starts from a student that behaves identically to the frozen model.

Contributions. **(1)** A latent knowledge-injection system for frozen instruction-tuned LLMs, evaluated with generative exact match on PopQA against text baselines and an untrained-injection control (§5.1). **(2)** Matched scaling measurements along three axes: token budget, training-corpus size, and frozen-model size, repeated across two model families (Qwen3 and Gemma-3), showing how the three trade against each other and which patterns are family-general (§5.2). **(3)** Causal probes showing that answers follow the injected graph, zero-shot transfer of the trained encoder to an unseen knowledge graph, and a cross-lingual study locating the transfer's boundary at the label language (§5.3, §5.4, §5.6). **(4)** An open release of the corpora, checkpoints, evaluation harness, and per-item outputs, enabling paired re-analysis of every reported comparison.

2 Related work

Lineage. The problem framing descends from work that probes language models as knowledge bases [45, 46] and its long-tail refinement [43, 42, 2], with RAG [44] as the deployed remedy. Knowledge-enhanced pretrained models such as KnowBERT [47], K-BERT [48], and K-Adapter [49] inject graph signal by modifying the model itself, which our frozen-model constraint rules out. ConceptFormer v1 [1] injected pretrained KG node embeddings (TransE-family vectors at PyTorch-BigGraph scale [50, 51]) through a per-entity lookup table. The present system builds its features from the frozen model's own label embeddings instead, which is what makes the encoder inductive. Prepending learned continuous tokens to a frozen model descends from prefix-tuning and soft prompts [52–54], textual inversion [55], and most directly from Frozen [56] and Flamingo [58]. Our substrate is Wikidata [59], and the verbalization baselines follow the graph-to-text encoding line [60].

Soft-token context compression. Gist tokens [26] and AutoCompressor [27] adapt the language model itself. ICAE [25] and 500xCompressor [28] train LoRA encoders against frozen decoders with reconstruction objectives, and CompLLM [30] distills per-segment embeddings through activation matching. ARC-Encoder [31] is a recent text compressor for frozen decoders. IC-Former [3] is architecturally similar: learnable digest tokens cross-attend over the LLM's word embeddings in a small standalone encoder. A parallel personalization line builds reusable per-user embeddings with resamplers for frozen LLMs [4, 5]; these are per-entity soft tokens in structure, without graph input and without label-free training. PISCO [29] follows the same principle as our objective, distillation from a model that reads the uncompressed text, but implements it with sequence-level distillation on teacher-generated answers and LoRA-tunes both compressor and decoder. It reports that a frozen decoder underperforms at document granularity; at entity granularity, a frozen decoder suffices (§5.1).

The most directly comparable system is xRAG [24], which injects into the input embeddings of a frozen decoder and includes a KL term against the same model reading the text. Its training additionally relies on paraphrase pretraining and about one million labeled instruction examples. Its representation is a single pre-existing retriever embedding passed through a projector, which binds it to that embedding model at inference. Its evaluation covers aggregate QA accuracy and robustness to noisy retrieval, and does not test unseen-entity generalization or the content of the compressed representation. Cartridges [32] distills each corpus into a trainable KV cache by gradient descent on self-generated conversations. This amortizes over the queries to one corpus, but there is no encoder, so a new corpus requires a fresh optimization run. The same teacher-reads-text principle also appears outside the input space: prompt distillation [6] tunes LoRA adapters toward the same model reading new documents, context distillation refines this in weight space [7], and KV-Distill [8] applies the KL objective to compress KV caches. A systematic study of gist-token compression [9] catalogs what such representations drop, including the counterfactual failures our probes target. These systems compress unstructured text and evaluate on task accuracy; none compresses structured input or probes the compressed representation causally. Post-hoc overflow probing [33] is the nearest attempt.

Knowledge-graph injection into LLMs. GNP [36] and G-Retriever [38] retrieve a question-specific subgraph and run a graph encoder over it at inference time, so every question pays a retrieval and encoding pass. Both train with next-token cross-entropy on labeled tasks, GNP with an auxiliary link-prediction loss, and G-Retriever also feeds the retrieved subgraph as plain text. LGPT [10] refines this family with learnable graph pooling tokens, still computed per question. GraphToken [37] encodes small synthetic graphs for graph-reasoning questions; its encoder never reads the question, but its graphs contain no entities, and it names factual grounding as future work. LLaGA [39] projects frozen text-encoder features of a node's sampled neighborhood into token space and is structurally the closest to per-entity encoding, but it rebuilds the sequence for every instance and targets node classification and link prediction rather than factual recall. All four keep the LLM frozen and train the encoder or projector on task labels. All four also recompute their graph representation for every question, so none produces reusable per-entity tokens. KBLaM [34] precomputes one key-value pair per triple with a sentence encoder and learned adapters, injecting them into every attention layer through a modified attention with retrained query heads. New triples need only an encoder pass, but training requires supervised QA pairs, and on the one real knowledge base it is tested on, built from Enron emails, its accuracy degrades. SR-KI [11] scales the same recipe to tens of thousands of triples with a supervised attention loss; its per-triple latents are reusable, but training requires labeled QA, and it injects flat triples where ours are compressed neighborhoods. ReaLM [12] quantizes pretrained KG embeddings into discrete tokens in the LLM vocabulary for knowledge-graph completion, which trains the vocabulary and targets a different task. Knowledge Prompts [35] learns one soft prompt per entity for 1.1M Wikidata entities as free parameters in a lookup table, so covering a new entity means training a fresh prompt; no encoder exists to produce one. Recent analyses find that graph-token LLMs "do not fully understand graph tokens" [40] and that graph tokens can decouple from graph semantics [41]. Our counterfactual evaluation (§5.3) measures this decoupling directly.

3 Method

Setup and featurizer. Let M be a frozen chat LLM with input-embedding dimension d. For an entity e with 1-hop neighborhood $G_e = \{(p_i, n_i)\}_{i=1}^{N}$ (property and neighbor labels, PageRank-ordered), each edge becomes $x_i = [\,\phi(p_i); \phi(n_i)\,] \in \mathbb{R}^{2d}$, where ϕ mean-pools M's input embeddings over a label's tokens. The features live in M's own embedding space, and the system holds no per-entity parameters.

Encoder. The encoder is a latent-query resampler [57, 58]. In each of L layers, k learned queries cross-attend over the edge set (order-invariant, padded edges masked), self-attend, and pass through a feed-forward block (d_{model}=1024, L=4, roughly 70M parameters; Table 5 ablates this capacity).

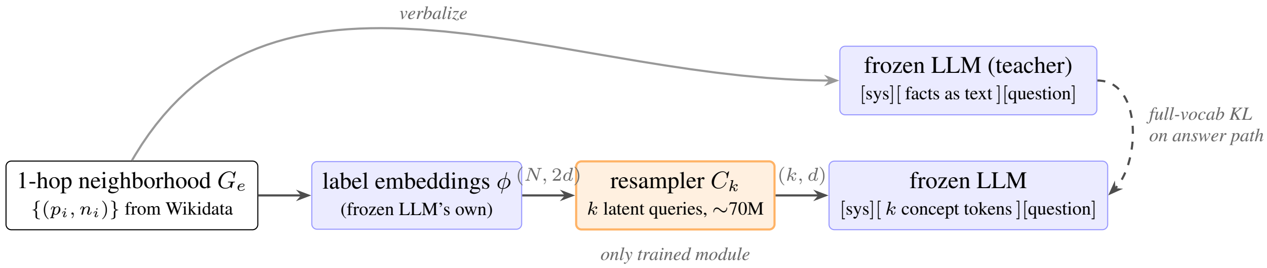

Figure 1: ConceptFormer-2. Edge features are the frozen model's mean-pooled label embeddings, a Perceiver-style resampler compresses the edge set into k concept tokens, and the tokens are spliced into the frozen model's input. The training signal is the KL between the same frozen model reading the facts as text (teacher) and reading the k tokens (student).

An output projection maps the queries to $\mathbb{R}^{k \times d}$. The projection is zero-initialized, so at step 0 the concept block has no effect and the student's next-token distribution matches the frozen model to within numerical noise. Because the encoder input is only the entity's neighborhood, the concept tokens are independent of the question: they can be computed once per entity and cached, and an unseen entity needs a single encoder forward pass.

Objective. The teacher is M reading [system][verbalized G_e][question], and its greedy answer path $y_{1..m}$ is stored offline. The student is M reading the same question, with the concept block $C_k(G_e)$ spliced in immediately before the entity mention at ordinary consecutive token positions. Table 5 ablates this placement; we anchor the block to the mention because a prompt with several entities can then carry one block per mention. The loss is the token-level full-vocabulary KL [62] along the stored path,

$$\mathcal{L} = \frac{1}{m} \sum_{t=1}^{m} \mathrm{KL}\big(p_M(\cdot \mid \text{facts}, q, y_{<t}) \,\|\, p_M(\cdot \mid C_k(G_e), q, y_{<t})\big)\,. \tag{1}$$

Only the roughly 20 path positions pass through the LM head, which cuts the per-step compute.[1] One converged run fits a single 24 GB GPU. The corpus questions (generated by a separate LLM, described below) serve only as inputs for the teacher. No QA labels enter the loss.

Data. From the Wikidata PageRank ranking we sample entity pools, excluding all PopQA subjects, and snapshot each entity's complete 1-hop neighborhood. A separate open-weights Gemma model generates questions from these neighborhoods. Each question is then tiered: it enters the corpus if the graph helps the backbone, meaning the base model fails the question and the model with the facts in context answers it (a 30% sample of base-known questions is retained). Finally, we store the teacher's greedy answer paths. The corpora hold 10k entities (92,177 rows under the Qwen3-0.6B [63] teacher) and 100k entities (925,178 rows). Tiering and teacher paths depend on the backbone, so each backbone trains on its own corpus variant of near-identical size, and cross-backbone comparisons evaluate each model with its own curriculum. The same pipeline produces the corpora for the six backbones of the two model families we study (Qwen3 0.6B/1.7B/4B and Gemma-3 270m/1b/4b [20]).

Training. AdamW with constant learning rate 10^{-4}, weight decay 0.01, and effective batch size 32 through gradient accumulation. We train 72k optimizer steps on the 10k corpus and 100k steps (about 4 epochs) on the 100k corpus, and select the best checkpoint on a held-out sample. Reported configurations use three random seeds unless a table states otherwise.

[1] In implementation the batch loss averages KL over all path tokens in the batch (a token-weighted mean); under gradient accumulation each micro-batch is weighted equally.

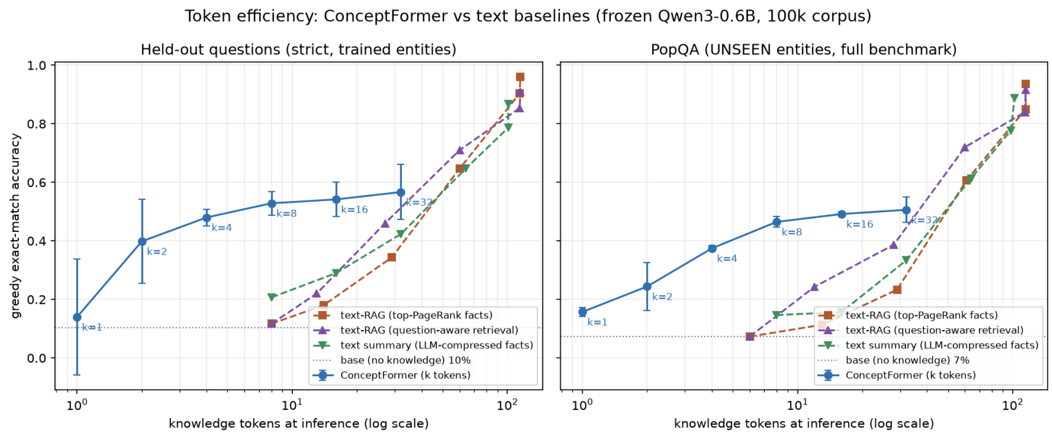

Figure 2: Accuracy against knowledge tokens spent at inference (log scale), frozen Qwen3-0.6B, 100k-entity corpus, 3 seeds. Concept tokens are compared with three text baselines at the same token budget: top-PageRank truncation, question-aware retrieval, and LLM-written summaries. All curves are scored on the same frozen 300-item eval subset per axis (the text baselines verbalize facts without an answer guarantee). Left: strict held-out. Right: PopQA.

4 Evaluation protocol

Every configuration is scored between two conditions of the same frozen model: **base**, the model without any knowledge, and **RAG**, the model reading the verbalized facts with the questioned edge guaranteed present. Base gives the floor and RAG the ceiling. Both are backbone-specific; on PopQA the ceiling is 0.96 for the primary Qwen3-0.6B backbone and stays above 0.82 across the other five. The metric is greedy exact match with word-boundary alias matching. We additionally report the official PopQA metric, which is more lenient and tracks ours within about 0.5 pt.

We report accuracy on three sets. **Held-out** questions are unseen questions about trained entities. **Held-in** questions repeat training questions and gauge overfitting. **PopQA** [61] asks about entities the encoder never saw and measures inductive generalization (n=14,266 questions with snapshot coverage). Held-out uses a question-level split with a post-hoc strict filter that removes every held-out question sharing an (entity, fact) pair with any training question (tables mark this leakage-free set as strict). The strict set is skewed toward facts asked only once, so held-out accuracy is measured on a subsample that is not representative of the full fact distribution. All systems are scored on identical item sets, which permits paired statistics: we report Wilson intervals and exact McNemar tests. Checkpoints are selected on a small held-out sample and scored on larger sets that contain it. Per-item outputs for every evaluation are released.

5 Results

5.1 Token efficiency against text baselines

Figure 2 compares concept tokens with three text baselines at matched knowledge-token budgets: query-independent top-PageRank truncation, question-aware retrieval that ranks facts by embedding similarity to the question, and LLM-written budgeted summaries. The question-aware baseline sees the question and can select the one relevant fact, an advantage the query-independent concept tokens do not have. On unseen entities at a budget of eight tokens, concept tokens reach 0.48 accuracy and the best text baseline reaches 0.15. The text baselines reach the same accuracy at roughly 36 to 49 knowledge tokens (a budget of 32 to 64), so matched accuracy costs text about five to six times more tokens. An untrained control separates the encoder from the injection mechanism: filling the same k slots with mean-pooled embeddings of the top-k edges lifts PopQA accuracy only from 0.103 to

Table 1: The k-family on the 100k-entity corpus (frozen Qwen3-0.6B, 3 seeds, mean$\pm$std). Strict held-out n=2000, PopQA n=14,266. The official PopQA metric is reported for comparability with the literature. Base scores 0.103; RAG, reading the verbalized facts with the answer edge present, scores 0.960 (an oracle-retrieval ceiling). Each adjacent k step is significant under an item-paired McNemar test ($p < 10^{-5}$).

k	1	2	4	8	16	32
held-out	.298±.006	.340±.023	.441±.007	.545±.014	.594±.024	.610±.011
PopQA	.203±.008	.298±.075	.412±.023	.477±.002	.518±.020	.537±.010
PopQA (official)	.207±.009	.303±.077	.419±.024	.483±.002	.523±.021	.543±.010

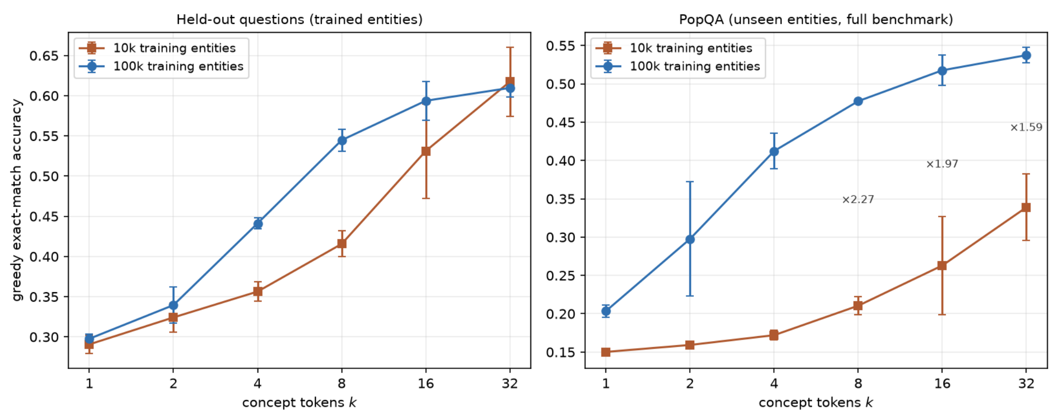

Figure 3: Token-budget curves at two corpus scales (frozen Qwen3-0.6B, before-entity placement throughout, 3 seeds per cell except the 10k k=8 cell with 6). Left: strict held-out. Right: PopQA, with the 10k→100k accuracy ratio annotated per k.

0.130. The untrained slots recover 7% of the trained gain, so the effect comes from the learned compression.

5.2 The three-way trade: tokens, data, and model size

Token axis. On the primary Qwen3-0.6B backbone, accuracy is monotone in k through k=32 on both corpora (Table 1, Figure 3). Returns per token diminish: k=8 reaches about 89% of the k=32 PopQA accuracy with a quarter of the tokens. The best budget therefore depends on how much context a deployment is willing to spend, and the curve shows no plateau up to k=32.

Data axis. Scaling training entities 10× at k=8 lifts PopQA accuracy from 0.210 to 0.477 on identical evaluation sets (Table 2). Held-out accuracy rises in step (0.416 → 0.545) and was still climbing when training ended, while held-in accuracy falls, consistent with the encoder memorizing individual training entities less and generalizing the edge-to-concept mapping more.

Token–data substitution. The two axes interact. The 10k→100k PopQA ratio rises from 1.35 at k=1 to about 2.3 at k=4–8, then falls to 1.58 at k=32 (annotations in Figure 3). Small token budgets cannot exploit more training data; at large budgets the extra tokens extract more value from the small corpus, so the token budget partially substitutes for training data. The high-k 10k cells carry wide seed spreads (±0.04–0.06), so the end of the curve is less settled than its middle.

Model axis. We repeat the data transect across the Qwen3 family, each backbone with its own teacher-relative corpus and brackets (Figure 4, Table 2). A larger frozen model knows more of PopQA from its weights, so the grid asks whether it gains less from injection and whether more training data closes the gap. It gains less, and data does not close the gap. The 1.7B margin is roughly half the 0.6B margin at both corpus scales, and the same 10× data step multiplies the 0.6B

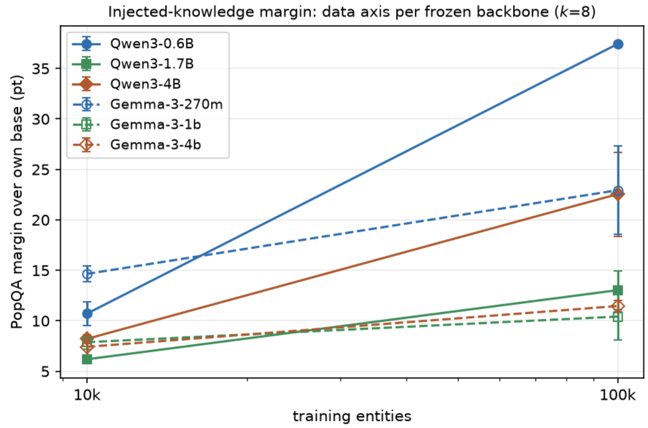

Figure 4: PopQA margin over each frozen backbone's own base accuracy at k=8, for both training-corpus sizes. Cell values and seed counts are given in Table 2.

Table 2: The data × model grid at k=8: full-PopQA accuracy and margin over each frozen backbone's own base (mean±std over seeds). Corpora are teacher-relative (§3), so each backbone trains on its own curriculum of near-identical size.

backbone	base	10k entities	100k entities	data gain	seeds
Qwen3-0.6B	.103	.210±.012 (+10.7)	.477±.002 (+37.4)	3.5× (margin)	6 / 3
Qwen3-1.7B	.141	.203±.003 (+6.2)	.271±.019 (+13.0)	2.1× (margin)	3 / 3
Qwen3-4B	.154	.237±.002 (+8.2)	.380±.042 (+22.5)	2.7× (margin)	3 / 3
Gemma-3-270m	.039	.185±.008 (+14.7)	.268±.044 (+22.9)	1.6× (margin)	3 / 3
Gemma-3-1b	.112	.191±.007 (+7.9)	.217±.023 (+10.4)	1.3× (margin)	3 / 3
Gemma-3-4b	.167	.241±.008 (+7.4)	.281±.006 (+11.5)	1.6× (margin)	3 / 3

margin far more than the 1.7B margin (data-gain column, Table 2). Headroom does not explain this: base accuracy moves only from 0.10 to 0.15 across the Qwen family and its RAG ceiling stays near 0.97, yet at 100k entities the 0.6B closes 44% of its base-to-RAG gap, the 4B closes 28%, and the 1.7B closes 16%.

Two further results bear on the model axis. The 1.7B held-out accuracy barely moves across the data step ($0.397 \to 0.414$) where the 0.6B gains 13 points. The encoder trains equally well on the bigger backbone, and what shrinks is specifically the payoff on unseen entities. The axis is also non-monotone: the 4B margin exceeds the 1.7B margin at both corpus scales (no seed-range overlap), so the 1.7B is a local minimum of the model axis and the 4B recovers. The teacher-relative curricula remain a candidate confound throughout, since a stronger base tiers a different question mix into its training corpus; for Qwen the tiering shifts monotonically with scale, so it cannot by itself produce the non-monotone valley.

A second model family tests how much of this is Qwen-specific. Across Gemma-3 backbones (Table 2), the margin again roughly halves from the smallest model to the mid-size one at both corpus scales, and the smallest backbone in either family posts the largest margin. This small-to-mid decline replicates cleanly. The recovery at the largest backbone does not: in Gemma the 4b exceeds the 1b only at 100k and within seed noise, and at 10k the family is monotone. The mid-size valley is therefore a clear effect in Qwen and, at most, a weak one in Gemma. Gemma's data gains are also uniformly smaller than Qwen's, so the value of additional training data itself depends on the family. Gemma-3-270m is a further caveat: its teacher-relative corpus is 16% smaller and its RAG ceiling is 0.82, not near 0.97, so its margin is not measured against the same ceiling as the larger Gemma backbones.

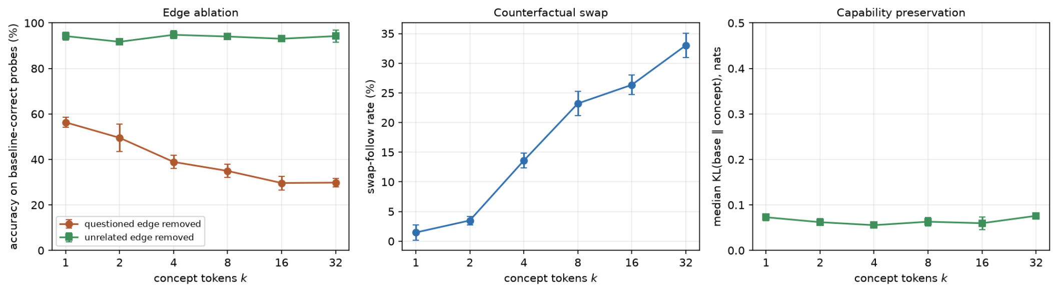

Figure 5: Causal probes across k (Qwen3-0.6B, 100k corpus, 400 single-fact probes per checkpoint, 3 seeds). Left: accuracy on baseline-correct questions after removing the questioned edge or an unrelated edge. Middle: rate at which the model follows a rewired edge to the false answer. Right: median next-token KL on off-topic control tasks between the model with and without concept tokens.

Stability across scale. Instability emerges at the larger corpus in both families, but on different backbones. Every 10k cell in either family has a seed spread within about 1.2 points. At 100k, the seed spread balloons on Qwen's two larger backbones and Gemma's two smaller ones, while Qwen3-0.6B and Gemma-3-4b stay within ± 0.6 (std column, Table 2). Run-to-run variance therefore tracks the specific backbone–corpus pair, not model size. Where variance is large it exceeds real effects: doubling the 1.7B's token budget at 100k lifts the mean margin to +15.1, less than the seed spread of either cell. Stabilizing the recipe per backbone is a prerequisite to measuring such effects at scale.

5.3 Causal probes of graph reading

A compression system can score well on QA while treating the graph as an opaque signal, a deficiency diagnosed across graph-token systems [40, 41]. Existing checks are correlational: diagnostic probes on resampler tokens in vision-language systems [15] and controlled edits of textual reasoning structures [16]. We instead intervene on the input graph of a trained model, re-encode it, and observe the answer.

Edge ablation removes single edges. Removing the questioned edge reduces accuracy on that question to 0.30 at $k{=}32$ (from 1.0 for baseline-correct probes; the residual is larger at small k), while removing an unrelated edge leaves accuracy at 0.94 (Figure 5, left). The dependence on the specific edge grows with k, and the residual reflects answers recoverable without that edge. The encoder stores edges largely separably.

The counterfactual swap rewires the questioned edge to a type-plausible false neighbor. A model that reads the graph follows the rewiring and produces the false answer. This swap-follow rate rises from about 1% at $k{=}1$ to 33% at $k{=}32$ (Figure 5, middle). The frozen base solves only 10% of these probes from its weights, so parametric memory cannot explain the behavior. The model reads the injected graph, and reads more of it as the token budget grows.

Injection perturbs the model only modestly on unrelated tasks. On control tasks that mention the entity without asking about its facts, the median next-token KL between the model with and without concept tokens stays between 0.05 and 0.08 nats, flat in k (Figure 5, right); the distribution is right-skewed, so a minority of controls shift more. Accuracy is also stable across prompt wording: across seven system prompts, PopQA accuracy of the $k{=}8$ model varies with a standard deviation of at most 1.1 pt.

5.4 Zero-shot cross-graph transfer

The featurizer consumes nothing but the frozen model's embeddings of label strings, so a trained encoder should in principle work on any labeled graph. Only an encoder that learned a general edge-to-concept mapping will actually do so. Transfer of this kind has precedent for graph retrievers,

Table 3: Zero-shot transfer to MetaQA-1hop (all 9,947 test questions, 3 seeds, mean±std). The Wikidata-trained checkpoints of Table 1 are evaluated without any MetaQA training. RAG verbalizes the same neighborhoods as text; since 1-hop neighborhoods fit the budget for over 99% of questions, it is effectively a full-neighborhood upper bound.

condition	accuracy
base (floor)	.058
k=1	.159 ±.002
k=2	.183 ±.018
k=4	.228 ±.015
k=8	.250 ±.034
k=16	.216 ±.039
k=32	.233 ±.088
RAG (reference)	.921

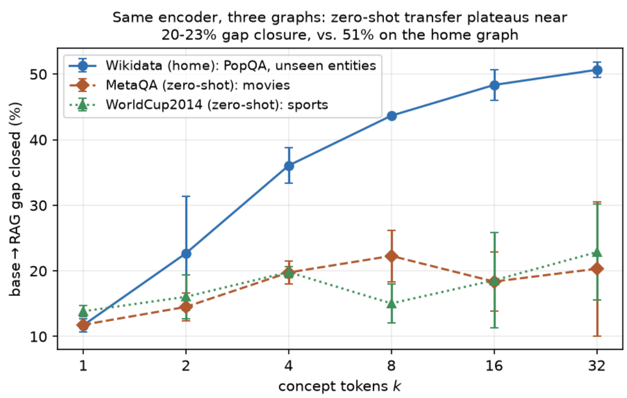

Figure 6: Share of the base-to-RAG gap closed vs. k: the home graph (Wikidata/PopQA) and two zero-shot foreign graphs, MetaQA (movies) and WorldCup2014 (sports). Three seeds each; bars are ±1 std. Zero-shot transfer tracks the home curve at small k but plateaus near 20–23% while the home graph climbs to 51%, so the returns to capacity above k≈4 are largely graph-specific.

whose graph foundation models generalize to unseen datasets while the LLM still reads text [13], and graph-token systems reach new domains through multi-domain instruction tuning [14]. Zero-shot transfer of the token interface itself has, to our knowledge, not been shown before. We test it on two foreign graphs. MetaQA [22] is a movie knowledge graph (43k entities, 9 relations) with a 1-hop QA benchmark in PopQA's style; WorldCup2014 [23] is a smaller football graph (1,088 entities, relations such as "plays in club" and "plays for country") with native 1-hop questions. Neither graph shares QA supervision with Wikidata, and the encoder is scored unchanged, so transfer on the task is zero-shot. The graphs are not fully disjoint from Wikidata: about 30% of their entity labels (the most famous entities) also occur in the training corpus, and for MetaQA two mapped relation labels ("cast member", "genre") coincide with Wikidata property labels; WorldCup's relation vocabulary is disjoint. We map each relation to a natural-language label (MetaQA needs hand-written reverse readings, e.g. `starred_actors` becomes "cast member"; WorldCup ships its own inverse edges); this map is a design choice, and transfer holds under a naive `underscore`-to-space mapping of the MetaQA relations (0.219 vs 0.250 accuracy at k=8, 3 seeds), so it is not an artifact of the hand-written labels. We encode each 1-hop neighborhood exactly as in §3 and score the Wikidata-trained checkpoints unchanged.

The concept tokens transfer to the unseen graph: they reach 0.25 accuracy at k=8 against a base of 0.058, clearing the base at every budget (Table 3). Figure 6 compares the two graphs on a common scale, as the share of the base-to-RAG gap that the concept tokens close. At small k the transferred and home curves coincide, and through k≈4 transfer captures roughly half to two-thirds of the home closure. Above that the two diverge: the home curve keeps climbing with k while the transferred

Table 4: In-domain adaptation on MetaQA-1hop (2,000-question eval split, 3 seeds, mean±std; the seeds are the three Wikidata initializations, and the adaptation run itself is deterministic). *zero-shot* is the Wikidata-trained checkpoint before any MetaQA training; *adapted* continues training the same encoder on MetaQA's own 1-hop questions with the label-free KL objective. No gold answer enters the loss.

k	zero-shot	adapted
1	.167 ±.003	.266 ±.007
2	.191 ±.015	.334 ±.043
4	.234 ±.013	.496 ±.037
8	.256 ±.037	.601 ±.010
16	.219 ±.040	.629 ±.011
32	.236 ±.088	**.658 ±.009**

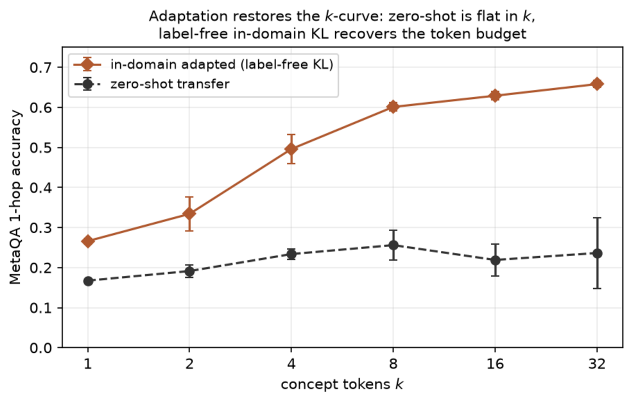

Figure 7: In-domain adaptation restores the k-curve on the foreign graph. Zero-shot transfer accuracy is nearly flat in k; a short label-free KL fine-tune on MetaQA's own questions recovers the returns to capacity, reaching 0.66 at k=32 against a 0.92 RAG ceiling. Three seeds each; bars are ±1 std.

curve plateaus and its seed variance widens. The extra capacity that helps on Wikidata does not survive the graph swap. The high RAG reference shows that the frozen model can answer in this domain when given text, so the remaining gap belongs to the transferred encoder. WorldCup shows the same shape on a second, unrelated domain: closure rises through $k \approx 4$, then flattens. The two foreign graphs land near the same endpoint, roughly 40–45% of the home-graph closure (Figure 6), which suggests the transfer ceiling is a property of the method rather than the target graph. We hold this reading loosely: the high-k cells are seed-fragile (one seed drives the k=32 mean), and WorldCup's questions cluster on a few hub entities, so its confidence intervals are optimistic. Together with the probes of §5.3, transfer supports the inductive claim: the encoder learned a mapping from labeled edges to concept tokens that generalizes beyond its training graph, though only its low-capacity part transfers without adaptation.

In-domain adaptation lifts accuracy well above the zero-shot result with the same label-free KL objective. Continuing to train the encoder on MetaQA's own 1-hop questions (distilling the frozen model reading the verbalized facts, with no QA labels in the loss) lifts every budget well above its zero-shot start (Table 4, Figure 7). The gain is more than a constant offset. Zero-shot, accuracy is nearly flat in k, around 0.2 across the sweep, as the extra concept tokens carry Wikidata-specific structure the movie graph cannot use. Adaptation recovers the token budget: adapted accuracy rises monotonically with k to 0.658 at k=32, against a 0.921 RAG ceiling, restoring the home-graph k-curve (Table 4). Adapting to a new graph is therefore cheap, and like the base objective it uses no QA labels in the loss. The transferred encoder is a useful initialization: a short fine-tune on the target graph's own questions closes most of the distance to text-RAG.

Table 5: Design ablations (10k corpus, k=8, 3 seeds each, strict held-out / PopQA). The augmentation test pools item-paired McNemar over 3 seeds; the effect is positive on average and one seed reverses on PopQA.

ablation	variants	held-out	note
encoder capacity	10M / 21M / **70M** / 231M	.423 / .441 / **.453** / .376	peak at ∼70M; largest is worst
placement	prefix / **before** / after / replace	.453 / **.423** / .434 / .380	**before** adopted (multi-entity)
prompt augmentation	off / **on**	.453 / **.472**	PopQA +2.0 pt; pooled p=3×10^{-36}
neighbor subsampling	**off** / on	$\Delta \approx 1$ pt	null; cached teacher is 2× faster

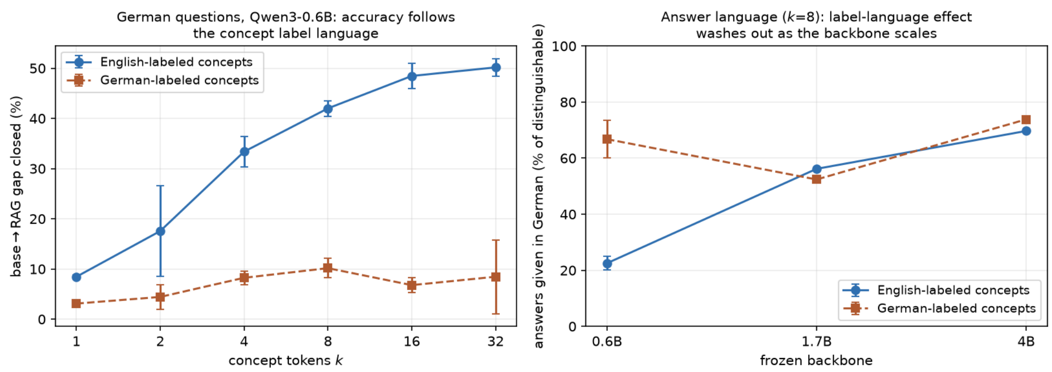

Figure 8: German PopQA (14,229 questions, German system prompt, localized entity mention). Left: on Qwen3-0.6B, base→RAG gap closure across k for concepts built from English vs. German entity labels. Right: at k=8, the share of correct answers given in German (of those where the German and English answer forms differ), by backbone.

5.5 Ablations

Encoder capacity peaks near 70M parameters (Table 5), and the largest encoder is worse at this training budget. Placement has little effect: prefix, before-entity, and after-entity cluster within noise, and only replacing the entity mention hurts. We adopt before-entity placement because it extends to prompts with several entities. Prompt augmentation gives a small gain that is significant under pooled paired testing but varies across seeds. Re-sampling teacher distractors at each step has no effect at our context budget.

5.6 Cross-lingual transfer

The encoder consumes only label embeddings, so cross-graph transfer (§5.4) raises the question of whether it also transfers across the label *language*. We translate PopQA into German with Gemma-3 [20] (14,229 questions), give the frozen model a German system prompt, and place the entity mention in its localized form (German where a localized label exists). Two conditions differ only in the concept tokens: built from the entities' English Wikidata labels, or from their German labels. Accuracy accepts an answer in either language, and we record which language each correct answer used. A system-prompt-language ablation (English vs. German) moved neither accuracy nor answer language, so we fix the system prompt to German.

The concept tokens' label language determines accuracy; the question and system language do not. English-labeled concepts recover 42–50% of the base-to-RAG gap on German questions at $k \geq 8$ (0.6B, 3 seeds), close to the English-question figure, while German-labeled concepts sit near the no-injection floor at every model size (Figure 8, left; Table 6). The encoder, trained on English label embeddings, transfers across graphs but not across the label language: German label embeddings are out-of-distribution for it, and the frozen model falls back to fluent but wrong German.

Table 6: German PopQA by concept label language (German system prompt, localized mention). Accuracy is base→RAG gap closure; "DE ans." is the fraction of correct answers given in German among those whose German and English forms differ. English-question home closure is shown for reference. The 0.6B rows are 3 seeds (mean±std); the 1.7B and 4B rows are a single seed.

	English-labeled concepts		German-labeled concepts		home (EN)
	closure	DE ans.	closure	DE ans.	closure
0.6B k=8	42±2%	22±3%	10±2%	67±7%	44%
0.6B k=32	50±2%	23±6%	9±7%	60±1%	51%
1.7B k=8	13%	56%	1%	52%	–
4B k=8	34%	70%	7%	74%	–

The label language does steer the *output* language, most strongly on the smallest backbone (0.6B, k=8; Table 6), where German-labeled concepts more than double the share of German answers. This effect narrows as the frozen model grows: at 4B both conditions answer in German about 70% of the time (Figure 8, right), because a stronger multilingual backbone follows the question's language on its own. Serving a new language with both accuracy and native output therefore needs either a capable multilingual backbone or an encoder trained on that language's labels.

6 Discussion

Ceiling or steerability. The model-size result admits two readings, and they make opposite predictions. Under a ceiling reading, a bigger base knows more of the benchmark, leaves less headroom, and more data recovers proportionally more of what remains. Under a steerability reading, a bigger frozen model needs more training signal per unit of behavioral change from k soft tokens, and the data gain itself shrinks with model size. Theory supports the second reading: soft prompts can bias a frozen model's attention outputs but cannot create new attention patterns, so they elicit rather than teach [17]. Empirically, prompt-tuning gains track base-model scale in fine-tuning scaling laws [18], and the marginal utility of retrieval also depends on model scale and pretraining saturation [19]. The completed Qwen grid rejects both readings as monotone accounts. Headroom is nearly constant across the family, which rules out a pure ceiling effect, and the injected margin is non-monotone in model size: it falls from the 0.6B to the 1.7B and recovers at the 4B (§5.2). The Gemma-3 family reproduces the small-to-mid decline; its recovery is within seed noise. Every configuration also shows large run-to-run spread at the larger corpus: in each family, at least one 100k cell has a seed range above 8 points, and which backbone is affected differs between the families. This spread is specific to entity generalization (held-out stays tight across the same seeds) and is measured under a fixed step budget, so we read it as an open obstacle to scaling injection rather than a settled bottleneck; the mid-scale valley is the least understood part. Per-backbone recipe tuning is the experiment that would decide it.

Limitations. Training uses a single knowledge graph and a single language, English Wikidata. Only entity-valued relations are modeled, so literals such as dates and quantities are absent. Evaluation centers on PopQA and on held-out questions from the same generator. The token-budget curves, faithfulness probes, and cross-graph transfer cover only the primary Qwen3-0.6B backbone, and the probes cover single-fact questions. Cross-graph evaluation is zero-shot on two benchmarks (3 seeds each), with in-domain adaptation shown on MetaQA; both are 1-hop. Several 100k cells carry wide seed spreads (ranges up to 8 points, on Qwen's larger backbones and Gemma's smaller ones), so their magnitudes are less settled than the stable cells. Corpora are teacher-relative, so cross-backbone comparisons involve each model's own curriculum. Knowledge is injected for one identified entity mention per prompt, multi-entity composition is untested, and neighborhoods are 1-hop.

Outlook. Several extensions are open. The encoder consumes embeddings, so multimodal knowledge such as entity images could enter through the same interface for natively multimodal frozen models, which is the subject of ongoing work. A multilingual encoder is a second direction: our encoder transfers across entities and across graphs, but it is trained on English label embeddings and does not carry over to labels in another language (§5.6), so serving a new language well will require training the encoder on that language's labels. A third direction turns the encoder into a graph embedder: representing each edge's neighbor by its own concept vector instead of its label embedding, and iterating, lets a single operator compose information across hops. In preliminary supervised experiments this recursion learns to answer 2- and 3-hop questions that single-hop concepts cannot, with warm-starting and in-domain fine-tuning both helping. A label-free version at scale remains to be tested. On the architecture side, injecting concept tokens into intermediate layers, in the spirit of deep prompt tuning [21], remains unexplored.

7 Conclusion

A frozen, instruction-tuned language model can be grounded in a knowledge graph through a handful of learned continuous tokens. Eight tokens more than quadruple accuracy on unseen entities at a fraction of retrieval's context cost. Causal probes indicate the model reads the injected graph, and the encoder transfers zero-shot to a graph it was never trained on. The injected margin shrinks as the frozen model grows, a decline that holds in two model families and is non-monotone in Qwen. Training stability at scale, together with this mid-size valley, is the main open question for scaling injection. All corpora, checkpoints, and per-item evaluation outputs are released.

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